

Methodologies for Software Processes

Seminar 2

Assignment 2

- Please complete the following tasks until Seminar 3.
- The Assignment 2 must be presented at the Seminar 3 (all group members must be in the class).

We will also use the SMT solver Z3, originally developed by Microsoft research:

<https://github.com/Z3Prover/z3>

Install Z3

Precompiled binaries for most operating systems are available here:

<https://github.com/Z3Prover/z3/releases/tag/z3-4.11.0>

Install Z3

To test your installation of Z3,
run the command line tool create a file **test.smt2**
with the following content:

```
(declare-const p Bool)
(assert (and p (not p)))
(check-sat)
```

Running **z3 test.smt2** should then output **unsat**.

Try and learn from the examples **ex-s1.smt2**, **ex-s2.smt2**, **ex-s3.smt2**, **ex-s4.smt2**, **ex-s5.smt2**, **ex-s6.smt2**, **ex-s7.smt2**, **ex-s8.smt2**, **ex-s9.smt2**, **ex-s10.smt2**, **ex-s11.smt2**

Install Python 3.12.2 from

<https://www.python.org/>

That includes also pip

<https://pip.pypa.io/en/stable/>

Install Z3Py Then install

C:\temp>pip install z3-solver

Try and learn from the examples ex-p1.py, ex-p2.py, ex-p3.py, ex-p4.py, ex-p5.py, ex-p6.py

Try this
example
ex-s2.stmt2

Example: Course Selection

- You have to take CS Modeling, Physics, or Chemistry
- For CS Modeling, you also need Discrete Math
- For Verification, you need CS Modeling
- For Physics and Chemistry, you need Statistics
- Statistics and Discrete Math are at the same time
- CS Modeling and Physics are at the same time
- Verification and Chemistry are at the same time

Is it possible to take Verification and all preliminaries?

Is it possible to take Physics and Discrete Math?

Exercise

- Model the following problem as an instance of the SAT problem.
- There are three chairs in a row: left, middle, right.
- Can we assign chairs to Alice, Bob, and Charlie such that:
 - Alice does not sit next to Charlie,
 - Alice does not sit on the leftmost chair, and
 - Bob does not sit to the right of Charlie?

The beginning of the solution is in `wedding.stm2` and `wedding.py`